FRANCK **ABOYA**

Passionate about software engineering, systems design, and applied AI, I specialize in **Java**, **Flutter**, **Python**, **SQL** and scalable backend architectures. Leveraging strong engineering fundamentals and technical precision, I build robust, production-ready software solutions; from cross-platform **mobile apps** and **REST APIs** to IoT-integrated systems and data-driven platforms; tailored to solve complex challenges in modern applications.

https://mesabo.github.io

INFORMATIONS

+81 070 9037 4702
mesabo18@gmail.com
Tokyo Koganei, Japan

LANGUAGES

Japanese: Beginner N4
English: Fluent
French: Native Spanish: Greetings

SKILLS

Abilities

- Rigorous
- Analytic
- Adaptable
- Crative
- Respectful
- Open mind

Skills

Programming Languages
- Java, Python, Dart, Bash, C

Artificial Intelligence
- Deep learning, LLM, IoT, Cloud dev

Databases
- SQL, NoSQL (MongoDB, Oracle)

Architecture & Dev
- APIs, Prototyping, Designing

INTERESTS

- Music
- Football
- Dance

WORK EXPERIENCES

FullStack Developer & AI

D'isum Inc.(Part-time, Tokyo) **Sep 2025 - Jun 2025**

- **Sensor Data Pipeline Architect & Analyst** (Emphasizes building pipelines from raw sensor data);
- **Sensor Data Specialist**: App Development, Prototyping & Integration (Covers full scope: development, architecture/prototypes, team collaboration on core software integration.)
- **Precision Sensor Data Engineer – Analytics, ML Prototypes & System Fusion** (Incorporates ML/predictive elements, prototyping, and integrating new systems with existing ones.)

AI Research and Development **Jul 2024**

Summer Internship –Japan AI Consulting, Osaka

- Worked with Generative AI, integrating Large Language Models (LLM) with backend API (Flask) and Database (log files + PostgreSql) using Cloud Dev.

FullStack Developer

Casys Technologies (Full-time) **Jan 2023 - Aug 2023**

- Improvement of backend systems in Java for commercial utility applications.
- Integrated backend systems with IoT devices such as vehicle sensors and door captors, enhancing system functionality.

BackEnd & FronEnd Developer

ANOHOU Solution (Part-time, Ivory Coast) **Au 2019 — Sep 2022**

- Developed systems , improving system performance and optimizing data management (Flutter, FastApi, NodeJs, GraphQL, MySQL, MongoDB).
- Member to the development of EKKOU app with ANOHOU Solutions, employing agile methodologies to deliver scalable solutions (Flutter)

PUBLICATIONS

Research Assistance at *YU Lab* - Tokyo (2025 – 2026)

Advancing Optimization through Lightweight LLM for IoT-enabled smart grids.

Research Machine Learning and Data Analysis (2024 – 2025)

International Collaboration for Enhancing Transformer-based models for improved performance in Energy Management solutions.

EDUCATION

Ph.D DC1 in Applied Informatics

Sep 2025 – Sep 2028

Hosei University – Tokyo, Japan

Master 2 in Applied Informatics

Sep 2023 – Sep 2025

Hosei University – Tokyo, Japan

INTERNATIONAL COLLABORATIONS

- ***TSFormer: Temporal-Aware Transformer for Multi-Horizon Forecasting with Learnable Positional Encodings and Attention Mechanisms,*** The 16th ICUFN 2025

- ***Federated Fine-Tuning of Large Language Models for Intelligent Automotive Systems with Low-Rank Adaptation,*** IEEE VTC2025-Spring.

- ***Dynamic IoT Resource Allocation Using Graph Reinforcement Learning,*** IEEE VTC2025-Spring.

- ***Enhancing Federated Learning in Consumer Electronics with Decoupled Knowledge Distillation against Data Poisoning,*** 2025 IEEE 22nd CCNC, doi: 10.1109/CCNC54725.2025.10976151.

- ***Enhancing Short-Term Load Forecasting in Internet of Things: A Hybrid Attention-based CNN-BiLSTM with Data Augmentation Approach,*** IEEE VTC2024-Fall.

- ***A Byzantine-Fault-Tolerant Federated Learning Method Using Tree-Decentralized Network and Knowledge Distillation for Internet of Vehicles,*** IEEE VTC2024-Fall.

- ***A Resource-efficient Text-to-Text Transfer Transformer Encoder-based Vertical Hybrid Model for Malicious URLs Detection,*** IEEE VTC2024-Fall.